

1. Algorithm

Main Algorithm

- Start the program.
- Display a menu containing the three string operations and an exit option.
- Read the user's choice.
- For choice 1, read a string and call the function that removes all vowels.
- For choice 2, read a string and call the function that finds the first repeating lowercase alphabet.
- For choice 3, read two strings and call the function that checks whether they are anagrams.
- Display the result returned or produced by the selected function.
- Repeat the menu until the user chooses Exit.
- Stop the program.

Algorithm for Each Function

A. Remove all vowels

- Traverse the string character by character.
- If the current character is not a vowel (a, e, i, o, u in either case), copy it to the output position.
- Add the string terminator at the end of the modified string.
- Display the modified string.

B. Find the first repeating lowercase alphabet

- Create a frequency array for the 26 lowercase letters and initialize all entries to zero.
- Count the occurrences of each lowercase letter in the string.
- Traverse the string from left to right.
- The first lowercase letter whose frequency is greater than one is the answer.
- If no such letter exists, report that there is no repeating lowercase alphabet.

C. Check whether two strings are anagrams

- Create a frequency array for all possible byte-character values and initialize it to zero.
- For each string, convert characters to lowercase and ignore spaces; update the frequency counts for the first string and subtract counts for the second.
- If every frequency is zero, the strings are anagrams; otherwise, they are not.
- Display the result.

2. C Code

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

#define SIZE 200

/* Removes vowels from the string in place. */
void removeVowels(char str[])
{
    int i, j = 0;

    for (i = 0; str[i] != '\0'; i++) {
        char ch = (char)tolower((unsigned char)str[i]);

        if (ch != 'a' && ch != 'e' && ch != 'i' &&
            ch != 'o' && ch != 'u') {
            str[j++] = str[i];
        }
    }
    str[j] = '\0';
}

/* Finds the first lowercase letter whose total frequency is > 1. */
char firstRepeatingLowercase(const char str[])
{
    int freq[26] = {0};
    int i;

    for (i = 0; str[i] != '\0'; i++) {
        if (str[i] >= 'a' && str[i] <= 'z') {
            freq[str[i] - 'a']++;
        }
    }

    for (i = 0; str[i] != '\0'; i++) {
        if (str[i] >= 'a' && str[i] <= 'z' &&
            freq[str[i] - 'a'] > 1) {
            return str[i];
        }
    }

    return '\0';
}

/* Checks anagrams, ignoring spaces and letter case. */
int areAnagrams(const char first[], const char second[])
{
    int freq[256] = {0};
    int i;
    unsigned char ch;

    for (i = 0; first[i] != '\0'; i++) {
        ch = (unsigned char)first[i];
        if (!isspace(ch)) {
            freq[(unsigned char)tolower(ch)]++;
        }
    }

    for (i = 0; second[i] != '\0'; i++) {
        ch = (unsigned char)second[i];
        if (!isspace(ch)) {
            freq[(unsigned char)tolower(ch)]--;
        }
    }

    for (i = 0; i < 256; i++) {
        if (freq[i] != 0) {
            return 0;
        }
    }

    return 1;
}

int main(void)
{
    int choice;
    char str[SIZE], str1[SIZE], str2[SIZE];
    char repeating;

    do {
        printf("\n===== STRING OPERATIONS MENU =====\n");
        printf("1. Remove all vowels from a string\n");
        printf("2. Find the first repeating lowercase alphabet\n");
        printf("3. Check whether two strings are anagrams\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");

        if (scanf("%d", &choice) != 1) {
            printf("Invalid input. Please enter a number.\n");
        }
    } while (choice < 4);
}
```

```

        return 1;
    }
    getchar(); /* Consume the newline after the menu choice. */

    switch (choice) {
        case 1:
            printf("Enter a string: ");
            if (fgets(str, SIZE, stdin) == NULL) return 1;
            str[strcspn(str, "\n")] = '\0';

            removeVowels(str);
            printf("String after removing vowels: %s\n", str);
            break;

        case 2:
            printf("Enter a string: ");
            if (fgets(str, SIZE, stdin) == NULL) return 1;
            str[strcspn(str, "\n")] = '\0';

            repeating = firstRepeatingLowercase(str);
            if (repeating != '\0') {
                printf("First repeating lowercase alphabet: %c\n",
                    repeating);
            } else {
                printf("No repeating lowercase alphabet found.\n");
            }
            break;

        case 3:
            printf("Enter the first string: ");
            if (fgets(str1, SIZE, stdin) == NULL) return 1;
            str1[strcspn(str1, "\n")] = '\0';

            printf("Enter the second string: ");
            if (fgets(str2, SIZE, stdin) == NULL) return 1;
            str2[strcspn(str2, "\n")] = '\0';

            if (areAnagrams(str1, str2)) {
                printf("The strings are anagrams.\n");
            } else {
                printf("The strings are not anagrams.\n");
            }
            break;

        case 4:
            printf("Exiting program.\n");
            break;

        default:
            printf("Invalid choice. Please choose 1 to 4.\n");
    }
} while (choice != 4);

return 0;
}

```

3. Result

The menu-driven C program was written using separate functions to remove vowels, find the first repeating lowercase alphabet, and check whether two strings are anagrams.

Sample outputs:

- Option 1 — Input: "Education" → Output: "dctn"
- Option 2 — Input: "programming" → Output: "r"
- Option 3 — Inputs: "Listen" and "Silent" → Output: "The strings are anagrams."

Note: The anagram function ignores spaces and letter case. The repeating-letter function considers lowercase letters a–z only and returns the earliest character in the string whose frequency is greater than one.